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Plant diversity within communities, not among them, stabilizes grassland productivity across spatial scales

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Evidence shows that local functional trait composition and diversity along the fast–slow leaf economics spectrum can predict the temporal stability of community productivity in response to environmental changes. However, it remains unclear whether these relationships persist at larger spatial scales. Combining a field survey of plant diversity with remote sensing estimates of primary productivity across a large environmental gradient in 235 grasslands across the Qinghai-Tibet Plateau and the Inner Mongolia Plateau, we find that species richness contributes to stabilizing productivity, while functional diversity contributes to destabilizing productivity at the local scale. In contrast, we find no evidence that variation in species richness and functional diversity among local communities contributes to stabilizing productivity across larger spatial scales. While the relationships between environmental conditions and local diversity and stability differ between the two regions, the overall positive relationships between diversity and stability are consistent at both the local and larger spatial scales. Our study offers insights into how functional traits along the fast-slow leaf economics spectrum mediate the effects of environmental factors (e.g., precipitation) on ecosystem stability at contrasting spatial scales.

Biodiversity loss, driven by climate change and human activities, is altering ecosystem functioning and diminishing the stable provisioning of essential ecosystem services[1,2]. There is a general consensus in grassland ecosystems that local species richness benefits community stability—the temporal stability of community biomass productivity[3–9] (hereafter 'community stability'). Beyond species richness, the trade-offs in functional traits along the fast–slow leaf economics spectrum have been shown to predict community stability[10,11] through two primary mechanisms[12]. First, resource acquisition strategies[13,14] determine species' responses to environmental changes, with fast species exhibiting faster recovery but lower resistance and slow species having higher resistance but slower recovery[10,13,15]. In this context, while communities dominated by fast species can exhibit high community stability due to high resilience, communities dominated by slow species can also exhibit high community stability due to high resistance[10,11] (Fig. 1a, c). A second mechanism is based on compensatory dynamics that result from trait (di)similarity: species can either respond asynchronously to environmental fluctuations when they differ in their functional traits (Fig. 1c), or display negative covariance due to competition if they have similar traits (Fig. 1a). Therefore, communities with high functional diversity in fast–slow traits are likely to show greater stability due to compensatory dynamics, while communities with low functional diversity may also be stabilized due to negative covariance[10,12,15].

Although most research on biodiversity–stability relationships comes from local-scale studies[7,9,16], there is growing interest in

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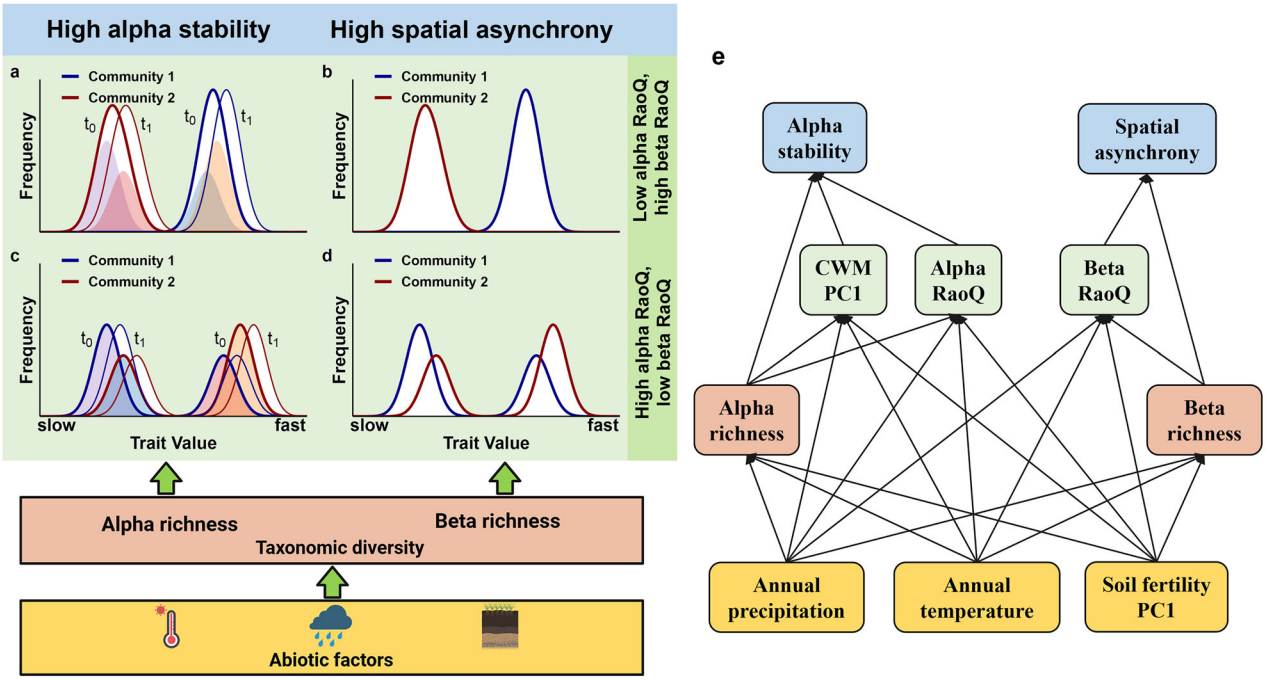


Fig. 1 | Conceptual figure illustrating how abiotic factors influence alpha stability and spatial asynchrony via taxonomic diversity, functional composition, and functional diversity across spatial scales. a, **c** Trait distributions of two species in two communities showing how different combinations of functional composition (community weighted mean of traits summarized by the first principal component, CWM PC1, positively related to fast strategy traits, including specific leaf area, leaf nitrogen content, and leaf phosphorus content) and alpha functional diversity (Rao's quadratic entropy, RaoQ) affect alpha stability. High alpha stability can arise when communities have low alpha RaoQ (**a**), or high alpha RaoQ (**c**), regardless of whether they are dominated by fast or slow strategy traits. Different colors represent different species within a community. Red and blue lines represent two different communities; bold lines indicate time t_0 and thin lines indicate time t_1. The *y*-axis represents trait frequency, which can be interpreted as the weighted abundance of a species in the community. with higher peaks indicating dominance. Horizontal distances between species reflect functional dissimilarity, while distribution width reflects functional diversity within or between communities. **b**, **d** Trait distributions of two communities in a metacommunity illustrating how functional diversity among communities (beta RaoQ) affects spatial asynchrony. High spatial asynchrony can emerge under both high beta RaoQ (**b**) and low beta RaoQ (**d**). Red and the blue lines represent the trait distributions of the two communities depicted in (**b**) and (**d**), respectively. **e** A priori piecewise structural equation model (SEM) summarizing hypothesized pathways linking abiotic factors, functional composition, and functional diversity to alpha stability and spatial asynchrony. Covariance paths and direct effects of abiotic factors on stability are omitted for clarity but are included in the SEM. CWM and alpha RaoQ measure functional composition and functional diversity related to fast strategy traits within communities, while beta RaoQ measures functional diversity among communities. Each panel illustrates one representative scenario to convey the underlying conceptual mechanisms. Abiotic icons were created in BioRender. Qiao, F. (2026) https://BioRender.com/t31idid.

understanding whether such relationships hold at larger spatial scales[17,18]. This shift reflects the growing need for managers to implement decisions at the landscape scale[17,19,20], where uncertainty from global environmental changes complicates outcomes. Advances in remote sensing technology[21–24] coupled with a theoretical framework that partitions stability across spatial scales[25,26] are enabling such investigations. Temporal stability of total productivity of communities at large spatial scales (gamma stability) can be partitioned into local community stability (alpha stability) and asynchronous dynamics of productivity across local communities (spatial asynchrony). The link between local diversity (alpha diversity) and alpha stability can be scaled up to the link between dissimilarity in species composition (beta diversity) and gamma stability. While local species richness (alpha richness) usually increases alpha and gamma stability[27,28], there is mixed evidence that dissimilarity in species composition among local communities (beta richness) contributes to spatial asynchrony and gamma stability[27–31].

Since functional trait diversity is a strong driver of diversity–stability relationships at the local scale[9–11,32], this importance may also extend to larger spatial scales. Integrating functional traits at local and larger spatial scales into the stability partitioning framework facilitates understanding their contributions to spatial asynchrony due to either compensatory dynamics of both fast and slow communities (Fig. 1b), or negative covariance between communities with similar functional traits (Fig. 1d). Although beta functional diversity can enhance gamma stability by promoting spatial asynchrony at late successional stages[33], it is unclear how functional diversity scales up to affect gamma stability in natural grasslands in response to environmental changes.

Two of the most idiosyncratic grassland ecosystems in China, the Inner Mongolia Plateau (IMP) and the Qinghai-Tibet Plateau (QTP), are facing the consequences of climate change[34,35]. Research indicates that diversity-functioning relationships in the IMP are precipitation-driven[36–39], but temperature-driven in the high-elevation QTP[40–42], which may lead to contrasting diversity–stability relationships. In this study, we perform a field survey of plant diversity combined with remote sensing estimates of primary productivity in 235 grassland sites across the two regions. We explore (1) the relationships between species richness, functional diversity and composition related to fast strategy (based on specific leaf area (SLA), leaf nitrogen content (N), and leaf phosphorus content (P)), and temporal stability of productivity, from local (individual communities) to larger (an assemblage of four spatially separated communities) spatial scales; (2) the direct and indirect effects (mediated by species richness and functional diversity and composition) of environmental factors (temperature, precipitation, soil fertility) on alpha stability and spatial asynchrony of

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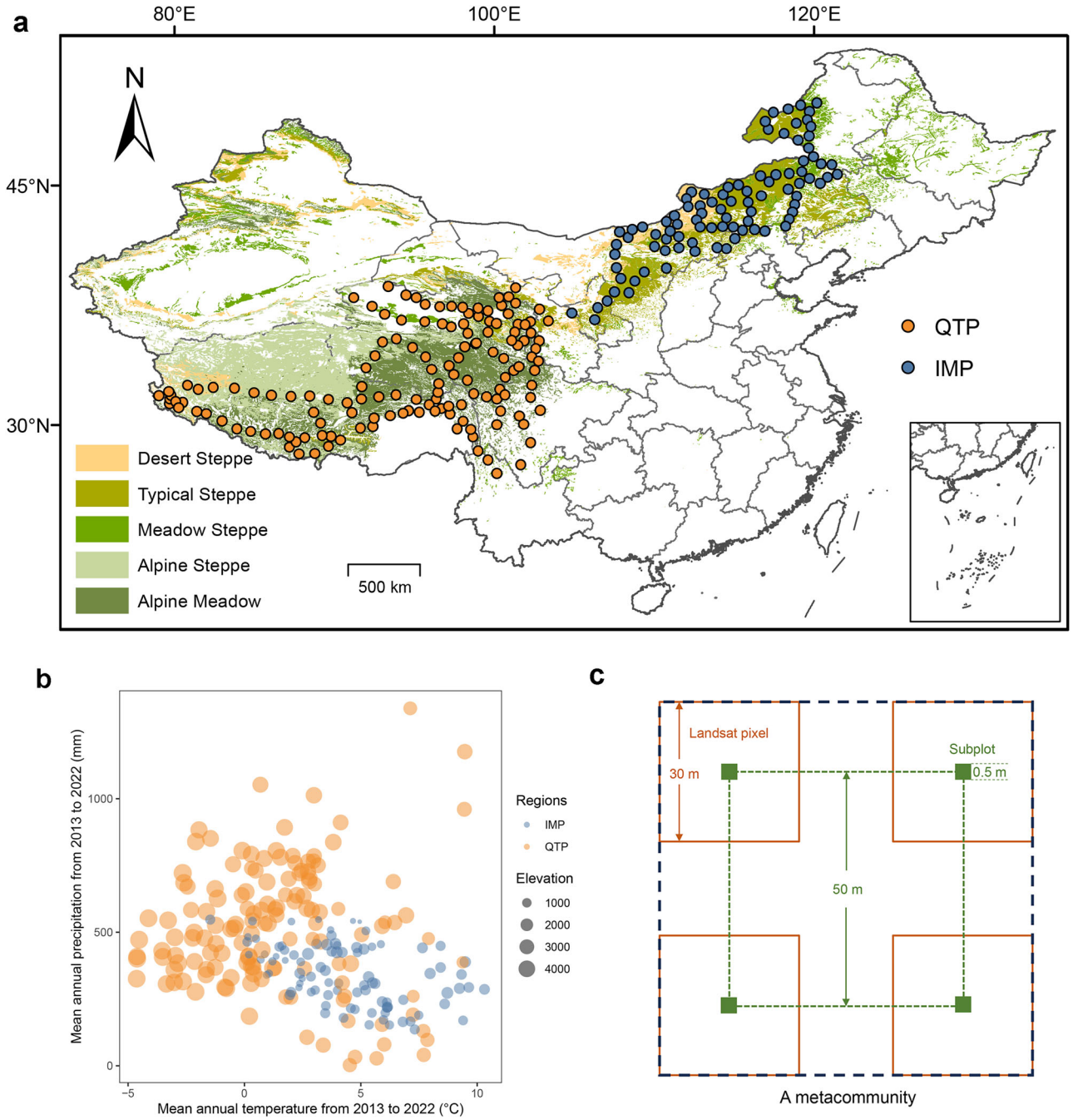


Fig. 2 | Study area, environmental gradients, and sampling design. a Locations of 235 sampling sites across the QTP and the IMP. **b** Relationships between mean annual temperature (MAT), mean annual precipitation (MAP) from 2013 to 2022, and elevation at each site. **c** Plot design (50 m × 50 m) at each site. Four subplots (0.5 m × 0.5 m) were selected at each site, and Landsat pixels (30 m × 30 m) were determined according to the latitude and longitude of each subplot. Details on the classification of grassland vegetation types are provided in Supplementary Method 3.

plant aboveground productivity (Fig. 1e); and (3) whether the pathways through which environmental factors affect stability differ between the two regions.

Results

Partial effects of diversity on stability across scales

Here, we assessed linkages between temporal stability of productivity, species richness, functional diversity and composition and environmental factors from local to larger spatial scales in 235 grassland sites across the QTP and the IMP in China (Fig. 2). Partial regression plots of the structural equation model (SEM; Fig. 1e) showed that, after controlling for the effect of other variables, both alpha richness ($p < 0.001$, $R^2 = 0.11$; Fig. 3a) and the first principal component derived from a principal component analysis (PCA) of community weighted mean of fast strategy traits (CWM PC1; $p = 0.003$, $R^2 = 0.04$; Fig. 3b) were positively related to alpha stability, while alpha functional diversity (measured as Rao's quadratic entropy, RaoQ; $p < 0.001$, $R^2 = 0.06$; Fig. 3c) was negatively related to alpha stability. Beta richness ($p = 0.099$, $R^2 = 0.01$; Fig. 3d) and beta RaoQ ($p = 0.693$, $R^2 < 0.01$; Fig. 3e) did not show a significant relationship with spatial asynchrony. These results suggest that different facets of alpha diversity contribute to local community stability,

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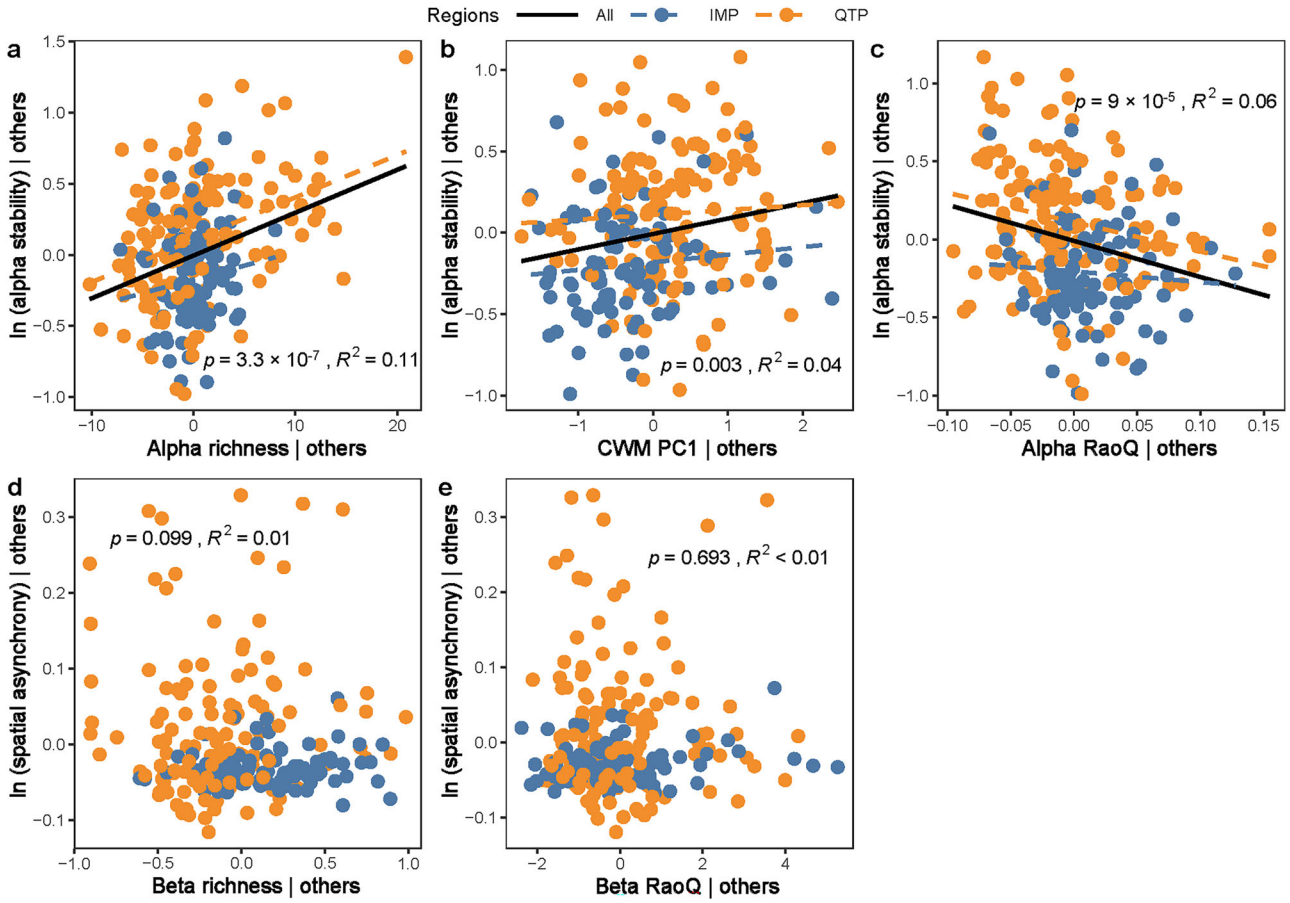


Fig. 3 | Partial regression relationships linking biodiversity to alpha stability and spatial asynchrony. Partial regression plots of SEM for the relationship between (**a**) alpha richness, **b** the first principal component of community weighted mean (CWM PC1) of SLA, leaf nitrogen (N) and leaf phosphorus (P) content and **c** alpha functional diversity (RaoQ) of aggregated specific leaf area, leaf nitrogen content and leaf phosphorus content and alpha stability after removing the effects of all other variables ("| others"), and between (**d**) beta richness and (**e**) beta functional diversity (RaoQ) of aggregated SLA, leaf N and leaf P and spatial asynchrony. Alpha stability and spatial asynchrony were log-transformed. Linear regression models were fitted with two-sided tests based on all sites across the IMP and the QTP. Regression lines are shown only when the regression across all sites is significant ($p < 0.05$).

whereas beta diversity does not contribute to larger-scale stability via spatial asynchrony.

Pathways through which environmental factors affect stability across scales and regions

Our multi-group SEM further confirmed that alpha diversity (both alpha richness and functional diversity) consistently contributed to alpha stability in both regions, while beta diversity (both beta richness and functional diversity) did not contribute to spatial asynchrony (Fig. 4). Specifically, the relationships between alpha diversity and alpha stability were robust across regions: alpha stability was positively related to alpha richness, negatively related to local functional diversity, and unrelated to the dominance of fast-growing species, as summarized by CWM PC1. Despite the negative relationships between alpha functional diversity and alpha stability, the overall relationships between alpha diversity and alpha stability remain positive. In contrast, the pathways through which environmental factors influenced stability at the two spatial scales differed between the two regions. At the local scale, environmental factors influenced alpha stability mostly indirectly via alpha richness and functional diversity, but they influenced spatial asynchrony directly at the larger spatial scale.

Specifically, at the IMP, mean annual precipitation (MAP) and soil fertility influenced alpha stability indirectly by increasing alpha richness and alpha functional diversity, while MAP also had a direct effect on alpha stability (Fig. 4a and Table S1). These combined direct and indirect pathways resulted in an overall positive relationship between both MAP and soil fertility with alpha stability (Fig. 4c). At the larger spatial scale, higher MAP directly promoted spatial asynchrony without influencing beta richness or beta RaoQ. Although a higher mean annual temperature (MAT) increased beta RaoQ, it did not further influence spatial asynchrony. Overall, MAP emerged as the strongest driver of both alpha stability and spatial asynchrony, mainly through a direct positive effect (Fig. 4c). Neither precipitation nor temperature variability contributed to alpha stability or spatial asynchrony (Fig. S1a).

Unlike the IMP, the direct positive contribution of MAP on alpha stability was not significant at the QTP (Fig. 4b). The previously reported positive effect of MAT on beta RaoQ was not observed, while a new negative effect on beta richness emerged. Alpha richness was the strongest driver of alpha stability, while MAP had the greatest influence on spatial asynchrony; both primarily through direct, positive effects (Fig. 4c). Additionally, increased precipitation variability reduced both alpha richness and alpha RaoQ, resulting in lower overall alpha stability (Fig. S1b).

Discussion

By combining a large-scale field survey of plant diversity with remote sensing estimates of climate and primary productivity across grasslands at the QTP and the IMP, our study reveals that alpha diversity,

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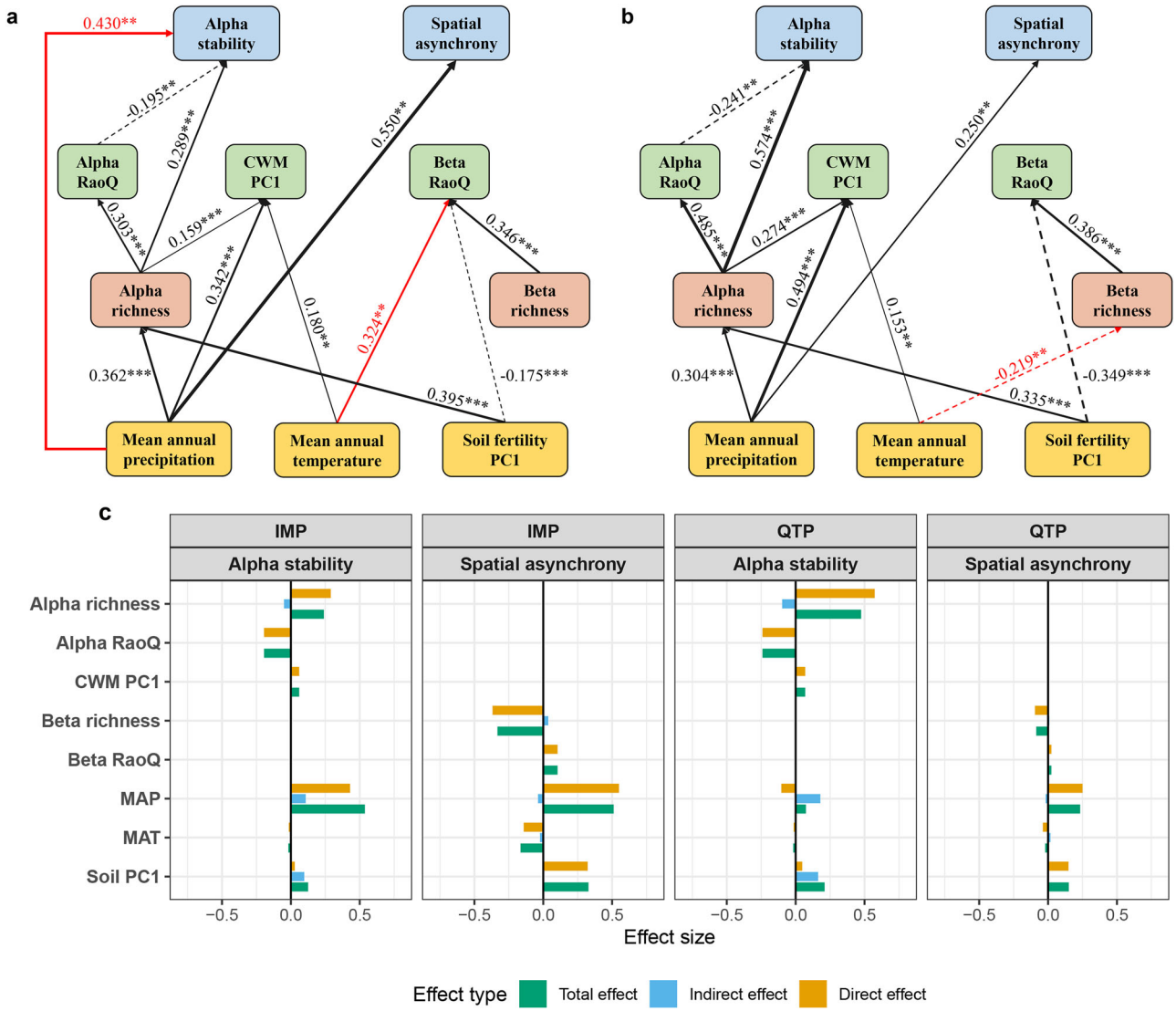


Fig. 4 | Multi-group SEM linking climate, soil fertility, and biodiversity to alpha stability and spatial asynchrony. Multi-group SEM showing pathways through which MAP, MAT, and soil fertility impact alpha stability and spatial asynchrony through taxonomic and functional diversity at different spatial scales in (**a**) the IMP and **b** the QTP. Black arrows represent significantly constrained paths ($p < 0.05$) while red arrows represent significant unconstrained paths between regions. Solid and dashed arrows represent positive and negative paths, respectively. Non-significant paths are not shown ($p \geq 0.05$). Numbers on arrows are standardized path coefficients (scaled by their mean and standard deviation), and asterisks indicate statistical significance ($^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$) based on two-sided tests. **a**, **b** Illustrate how pathways differ between regions, while (**c**) summarizes the direct, indirect, and total effects of predictors on alpha stability and spatial asynchrony for both regions, based on standardized path coefficients from the multi-group SEM. Soil fertility PC1, the first axis of aggregated soil fertility; CWM PC1, the first axis of the fast–slow functional composition; RaoQ, aggregated functional diversity of specific leaf area, leaf nitrogen content and leaf phosphorus content; MAP, mean annual precipitation; MAT, mean annual temperature. Overall model fit: Fisher's $C = 13.791$, $p = 0.314$, AIC = 113.790, d.f. = 12.

rather than beta diversity, influences ecosystem stability at large spatial scales. Specifically, species richness contributes to stabilizing productivity, while functional diversity related to the fast–slow leaf economics spectrum contributes to destabilizing productivity at the local scale. Additionally, whereas environmental factors influence alpha stability directly and/or indirectly by altering alpha-scale taxonomic and functional diversity, their influences on spatial asynchrony are direct; i.e., not mediated by changes in beta diversity. Although the environment modulated beta diversity and alpha stability differently in each region, the positive relationships between alpha diversity and alpha stability remained consistent. Our study demonstrates how alpha richness and functional diversity along the fast–slow leaf economics spectrum mediate the effects of environmental change on ecosystem stability across contrasting spatial scales.

While a number of studies have investigated the relationship between functional diversity and ecosystem stability across large environmental gradients[21,43,44], very few have explored the mechanisms linking functional diversity and stability beyond single local communities[33,45]. Our study combined these two approaches and found that, after controlling for the effects of other variables, alpha stability was positively related to CWM PC1, representing fast strategy traits, but negatively to alpha functional diversity (Fig. 4). Therefore, communities dominated by fast-growing species and exhibiting lower functional diversity were more stable. These findings contrast with previous studies, which showed that communities that were either dominated by conservative, slow-growing species[10,11,43], or that exhibited higher functional diversity, enhanced ecosystem stability in grasslands[46]. Moreover, our results also differ from a recent grassland

study that found no relationship between functional diversity and stability at local scales[33].

The relationship between the fast–slow leaf economics spectrum and community stability may be context-dependent[10]. Across our study regions, we identified five main grassland vegetation types: alpine meadow, alpine steppe, meadow steppe, typical steppe, and desert steppe, based on hydrothermal conditions, major climatic zones, and topographic features (Supplementary Methods 3). We observed a positive association between CWM of fast strategy traits and alpha stability in the meadow steppe and the alpine meadow, with relatively high precipitation in each region, and a negative trend in the desert steppe, which has low precipitation (Fig. S2f). This suggests that fast-growing species promote community stability due to fast recovery in frequently disturbed, high-precipitation sites, while slow species promote community stability through greater resistance to chronic environmental stresses in low-precipitation sites. However, this pattern was not observed after controlling for other potential confounding factors, such as precipitation. Instead, the overall pattern across vegetation types (Fig. S2b) and regions (Fig. 3b) showed a positive relationship between functional composition along the fast–slow leaf economics spectrum and community stability.

At the local scale, there was no difference between the two regions in the relationships of alpha richness or functional diversity with alpha stability (Fig. 4a, b). These results were similar for most vegetation types, except for the desert steppe (Fig. S2), where the relationships between alpha richness (Fig. S2a) or alpha functional diversity (Fig. S2c) and alpha stability were significantly negative. This aligns with previously reported negative diversity–stability relationships under high aridity conditions[47], probably resulting from decreased species asynchrony, as species tend to respond similarly to anomalous dry conditions, or increased temporal variability of community cover with increased species richness[47]. However, this finding contrasts with the positive relationship between species richness and stability under high aridity observed by García-Palacios et al.[21]. This discrepancy may stem from differences in spatial scales and vegetation types: while García-Palacios et al.[21] analyzed broad-scale patterns across global drylands, our study focuses specifically on desert steppe, where extreme abiotic stress and vegetation sparsity may override the stabilizing effects of diversity.

Although diversity–stability relationships were consistent at the local scale, the variables with the strongest total effects on alpha stability differed between the two regions (Fig. 4c). At the IMP, MAP was the strongest driver of alpha stability, which aligns with previous studies on community stability in that area[36,37]. This is likely because precipitation is the limiting factor in the area, and higher precipitation stabilizes biomass production[37]. In contrast, alpha richness was the strongest driver of alpha stability at the QTP. This finding is consistent with previous studies using remote sensing across heterogeneous Swiss landscapes[22] and in Dutch dune grasslands[48], but contrasts with studies showing that ecosystem stability depends more on environmental conditions in global drylands[21] and on the Tibetan Plateau[43]. The relative importance of diversity versus environmental conditions for ecosystem stability may therefore depend on ecosystem type and diversity[43]. In our study, the QTP had locally higher species richness (mean = 11, se = 0.68; Fig. S3a) than the IMP (mean = 7, se = 0.34) or the regions analyzed in Ren et al.[43] (mean = 8). Alpha richness is a stronger predictor of alpha stability than environmental conditions at the QTP, probably because higher species richness enables stronger compensatory dynamics and potentially greater functional redundancy, both of which stabilize community biomass.

At the larger spatial scale, no relationship was detected between beta richness or functional diversity and spatial asynchrony in either region. Although spatial asynchrony varied more at the QTP than at the IMP (Fig. 3d, e), no relationship was found even when analyses were restricted to the same ranges of temperature and precipitation in both regions (Fig. S4). There are a few factors that may have influenced the observed patterns. First, the lack of a relationship between beta diversity and spatial asynchrony could be due to the relatively homogeneous environments[28,31,49]. In contrast, environments with larger abiotic heterogeneity have shown positive relationships between beta richness and spatial asynchrony[17,45], as well as between beta functional diversity and spatial asynchrony[33,45]. Hautier et al.[27] found that beta diversity was positively associated with spatial asynchrony in a globally distributed network of natural grassland communities, but this relationship was not observed following fertilization, which reduced abiotic heterogeneity. Second, unmeasured functional traits may also help explain the neutral relationship between beta diversity and spatial asynchrony. For instance, belowground traits, particularly those related to fine roots, may have disproportionate effects compared to aboveground traits, as species with similar fine-root syndromes can exhibit contrasting aboveground traits[50]. In precipitation-driven systems, belowground traits (e.g., root depth) can respond to drought at rates and magnitudes that differ markedly from those of aboveground traits such as SLA. This response diversity across heterogeneous environments can enhance spatial asynchrony among communities, thereby strengthening the relationship between beta functional diversity and ecosystem stability. However, such belowground traits were not included in this study and remain underrepresented in large-scale assessments of ecosystem stability.

Third, the relatively short distance between our subplots could have led to high similarity in species composition, potentially limiting the detection of strong beta diversity and spatial asynchrony patterns. Synthesis studies have shown that seed dispersal distances for herbs and graminoids can reach up to several hundred meters[51,52], with a steep decline in dispersal likelihood probability as distance increases[51]. Given that the centers of our subplots were 50 m apart, this may represent a relatively small spatial scale compared with the dispersal capabilities of many species. Finally, the scale differences between field and remote sensing data may also influence the observed relationships between beta diversity and spatial asynchrony. Although previous studies have demonstrated that such mismatches do not necessarily compromise results[21,24], our field subplots were much smaller (0.5 m × 0.5 m) than the remote-sensing pixels (30 m × 30 m), which may amplify scale mismatch because finer-scale field measurements are more sensitive to small-scale heterogeneity.

Higher MAP increased alpha stability in both regions, but through contrasting pathways. At the IMP, precipitation increased alpha stability both directly and indirectly by increasing alpha diversity. In contrast, its direct effect was not observed in the QTP, weakening the positive relationship with alpha stability (Fig. 4c). This again highlights precipitation as a dominant factor in the IMP, whereas at the QTP, the direct effect of precipitation may have been masked by the influence of alpha diversity. Although higher MAP and temperature favored the dominance of fast-growing species, this did not translate into higher alpha stability, in contrast to the partial regression plots, where fast-growing species dominance was positively linked to alpha stability (Fig. 3b). Instead, greater precipitation led to reduced alpha stability, mediated by increased alpha functional diversity. This suggests that while dominant species' traits may contribute to stability, their effect was overshadowed by the stronger, negative influence of alpha functional diversity. This negative relationship may stem from low functional diversity promoting stability through negative covariance among similar, competing species, whereas high functional diversity may reduce stability due to a lack of functional redundancy.

In conclusion, our results demonstrate that while species richness and functional diversity contribute to alpha stability, they show no relationship with spatial asynchrony. The influence of environmental conditions on diversity and alpha stability differs between regions, yet the relationship between alpha diversity and stability is universally consistent. This suggests that management strategies aimed at

maintaining local diversity can reliably support ecosystem stability across different environmental contexts. Notably, the primary predictors of alpha stability differ between the two regions: MAP is the strongest driver in the IMP, whereas alpha richness plays the most prominent role in the QTP. The significance of our study lies in scaling up the relationship between functional diversity and ecosystem stability from local to larger spatial scales, thereby illustrating how functional diversity modulates the effects of environmental change on ecosystem stability across multiple spatial scales.

Methods

Study sites and sampling

We surveyed 43 grassland sites in 2021 and 201 sites in 2022 across the QTP (93 sites) and the IMP (151 sites), China, during the peak growing season (July and August). Our sites span a latitudinal gradient from 26.98°N to 50.15°N and a longitudinal gradient from 79.03°E to 121.50°E (235 sites in total after removing 9 sites; See below for details. Fig. 2a), with elevation ranging from 406.8 to 4957.0 m, and MAT and MAP ranging from −4.6 to 10.3 °C and from 2.7 to 1338.5 mm in the period between 2013 and 2022 (Fig. 2b). All sampling sites were located away from settlements, roads, and showed no signs of livestock activity to minimize the impact of human disturbance and ongoing grazing. At each site, a 50 m × 50 m plot was established, and 4 subplots of 0.5 m × 0.5 m in size were selected at the four corners of the plot (Fig. 2c). We recorded the latitude, longitude, and elevation of each subplot, and sampled species richness and aboveground biomass of the 4 subplots. Aboveground biomass was collected by harvesting all vegetation at ground level within each of the four systematically placed 0.5 m × 0.5 m quadrats per site. The harvested biomass was then sorted by species, dried at 65 °C for 72 h, and the species-level biomass was recorded.

Our study shows that 10.2% of the total species pool is shared between the two regions. We measured SLA for plant species whose accumulative biomass accounted for an average of 89.8% of the total biomass across all sites at the QTP and the IMP (Fig. S5a), ensuring that our functional traits reflect the dominant biomass-contributing species. We sampled five healthy, mature leaves from up to five randomly chosen individuals per species and site. Leaves were not detached from the plants in the field until just before scanning. Before that, individual plants were temporarily stored in moist paper envelopes, which were then placed in sealed plastic bags to retain humidity. All samples were kept in a cooler with ice packs immediately after collection and transported to the lab[53]. We scanned the leaves on the day of collection (within ~12 h) and measured leaf area from the scanned images using the software Digimizer (v.4.6.1; MedCalc Software Ltd., Ostend, Belgium), then dried the leaves at 65 °C for 72 h and measured the dry mass to a resolution of 0.0001 g. SLA was calculated as leaf area divided by its dry mass ($m^2\ kg^{-1}$)[53]. We also measured leaf N and leaf P at each site using the same leaves from SLA measurement in 72% of sites, subject to sample availability and species selection feasibility (the species for which we measured leaf nitrogen and phosphorus content accounted for an average of 81.5% of the total aboveground biomass across all sites; Fig. S5b). Leaf nitrogen content was measured by an elemental analyzer (Elementar vario MACRO cube, Germany). Leaf phosphorus content was measured using molybdenum-antimony resistance colorimetry with acid digestion on a Chemistry Analyzer (Smart Chem 450 Automatic Discrete Chemistry Analyzer, Smart Chem Technologies Ltd, China).

At each site, we collected six soil cores (3.8 cm in diameter and 15 cm in depth), combined them, and analyzed at the laboratory for their soil total carbon (STC; mg kg^{-1}), total nitrogen (STN; mg kg^{-1}), and total phosphorus (STP; mg kg^{-1}). STN and STC were measured using an elemental analyzer (Elementar vario MACRO cube, Germany). STP was determined by acid digestion (molybdenum-antimony resistance colorimetry) using Smart Chem 450 Automatic Discrete Chemistry Analyzer (SmartChem Technologies Ltd, China).

Satellite-based temporal stability of productivity and climate

We used the normalized difference vegetation index (NDVI) as a proxy for plant aboveground productivity. We also conducted the same analyses using NIRvP (Near-Infrared Reflectance of vegetation × photosynthetically active radiation), which yielded qualitatively similar results (Supplementary Methods 1). Therefore, we present only the NDVI-based results in the main text for simplicity. Field aboveground biomass collected in 2021 and 2022 at the subplot level showed a strong correlation with the corresponding NDVI data in 2021 and 2022, respectively (Fig. S6a, b). Specifically, we used data from the NASA Landsat satellite mission, which provides imaging data with a long time series and a pixel size of 30 m × 30 m over the region[54]. NDVI data for each subplot were acquired according to their latitude and longitude via the Google Earth Engine (GEE) platform from two Landsat missions (7 and 8) and Landsat Collection 2 Tier 1 Level 2 products to evaluate time series dynamics of plant productivity over a 10-year period from 2013 to 2022[55]. Each image was pre-processed by converting low-quality data (e.g., due to the presence of clouds, snow, or shadows) into missing values based on the masks provided with the downloaded images. We excluded 9 sites in the data analyses because at least two subplots are in the same pixel and have the same NDVI value due to insufficient spatial distance from each other (Fig. S7). We adopted the method of Oehri et al.[22] to smooth NDVI time series using a modified implementation of the harmonic analysis of time series algorithm (HANTS; Fig. S8). $NDVI_{ratio}$ method was used to determine growing season start (SOS) and end (EOS) from remote sensing data of each year[56]. SOS and EOS were defined as the first day of the year at which NDVI first exceeded and fell below the mean of its annual minimum and maximum value, respectively[22]. The average NDVI in the SOS and EOS time span was estimated and used as a proxy of the primary productivity of the year.

We also collected monthly mean air temperature and total precipitation data from 2013 to 2022 from NASA Famine Early Warning Systems Network Land Data Assimilation System (FLDAS) with a spatial resolution of 0.1 × 0.1 degrees for each site according to their latitude and longitude[57,58]. Then we calculated the mean and interannual variability (coefficient of variation, CV = SD/mean) of annual mean temperature and total annual precipitation from 2013 to 2022 for each site.

Species richness and stability across spatial scales

At each site, we treated each subplot as a community and the collective four subplots as a metacommunity[27]. We quantified alpha richness and temporal alpha stability at the community level, and beta richness and gamma stability across the four subplots at the metacommunity level. The community-level alpha richness for each site was the average species richness across the four subplots. We used the total number of species found in the four subplots as the gamma richness of each site, and beta richness was calculated by multiplicative partitioning[12,59].

For each site, we defined temporal stability of productivity at both scales as the ratio of the annual mean NDVI from 2013 to 2022 (mean NDVI) to the standard deviation of the annual NDVI (SD of NDVI) over this period[60]. Alpha stability of subplots was defined as the weighted average of community stability across subplots, weighted by the mean total biomass of each subplot. Gamma stability of the metacommunities was defined as the temporal invariability of total metacommunity biomass of the four subplots. Spatial asynchrony was defined as the asynchrony of total community biomass among local communities within a metacommunity[17,27]. The formulas behind these definitions are

described as follows:

$$Alpha\ stability = \frac{\sum_k \mu_k}{\sum_k \sqrt{v_{kk}}} \quad (1)$$

$$Gamma\ stability = \frac{\sum_k \mu_k}{\sqrt{\sum_{k,l} v_{kl}}} \quad (2)$$

$$Spatial\ asynchrony = \frac{Gamma\ stability}{Alpha\ stability} = \frac{\sum_k \sqrt{v_{kk}}}{\sqrt{\sum_{k,l} v_{kl}}} \quad (3)$$

Where μ_k and v_{kk} are the temporal mean and variance of NDVI of subplot k, respectively. v_{kl} is the covariance of NDVI between subplots k and l. Because gamma stability is totally explained by alpha stability and spatial asynchrony, we only showed the results of alpha stability and spatial asynchrony.

Functional trait composition and diversity, and soil fertility

For the calculation of functional diversity, we calculated the mean trait value of each species across all sites and retained only species with complete SLA, leaf N, and leaf P data. After this procedure, the selected species accounted for an average of 94.3% of the total aboveground biomass per site (Fig. S5c). To quantify the effects of dominant species' traits, we calculated the community weighted mean (CWM; calculated as the average of species trait values weighted by their relative biomass in the community) of SLA, leaf N, and leaf P in each subplot using the dbFD function in the *FD* package[61]. We used the first axis of a PCA of CWM of SLA, leaf N, and leaf P to represent the fast–slow leaf economics spectrum[10,13]. PCA was performed using the prcomp function in the *factoextra* package[62]. The first principal component of the PCA based on CWM (CWM PC1) captured 57.6% of the total variation among communities, with higher values indicating that communities are dominated by fast-growing species (Fig. S9). We calculated the average CWM PC1 across the four subplots in a site as CWM values at the community level. To quantify compensatory dynamics, we calculated functional diversity as Rao's quadratic entropy[63] (RaoQ) at alpha, beta, and gamma levels of the aggregated SLA, leaf N, and leaf P by the multiplicative partitioning method using the RaoRel function in the *cati* package[64,65]. We applied the Euclidean distance to calculate species dissimilarities with these three traits. We used "unweighted" species relative abundance, where each species is given equal weight, and the Jost correction proposed by de Bello et al.[66].

We also used a PCA based on STC, STN, and STP, and took the first principal component to represent soil fertility. The PC1 captured 83.2% of the total variation in soil nutrient concentrations, with a higher value of PC1 indicating higher soil fertility (Fig. S10).

Statistical analyses

We examined Pearson correlations among environmental variables (temperature, precipitation, and their variability, and soil fertility), diversity metrics, and stability indices to assess potential multicollinearity. No strong correlations were detected among the candidate variables (Pearson correlation coefficient $|r| > 0.7$; Fig. S11). We then constructed SEM with generalized least squares (GLS) regression models to assess the pathways by which MAT and MAP (or variability of temperature and precipitation) and soil fertility impact alpha stability and spatial asynchrony via both alpha (including alpha richness, CWM PC1, and alpha RaoQ) and beta diversity (including beta richness and beta RaoQ). First, we built a priori theoretical SEM reflecting hypothesized relationships among all variables (Fig. 1e; Supplementary Methods 2). Specifically, we related alpha richness, CWM PC1, and alpha RaoQ to alpha stability at the community level, and related beta richness and beta RaoQ to spatial asynchrony at the metacommunity level. To account for the potential effect of spatial autocorrelation (the Moran's I value for alpha stability and spatial asynchrony was 10.63 ($p < 0.001$) and 7.35 ($p < 0.001$), respectively), we used GLS models including a rational quadratic spatial correlation structure (corRatio) after comparing different autocorrelation structures by Akaike's information criterion (AIC) values (Table S2). We included the strongly correlated errors (e.g., X % ~ ~ % Y) based on the directed separation (d-sep) approach implied by the initial model to improve model fit. The goodness-of-fit test of the models was assessed by Fisher's C and p value ($p > 0.05$).

Based on the SEM we obtained above, we estimated the partial effects of alpha richness, CWM PC1, and alpha RaoQ on alpha stability, and the partial effect of beta richness and beta RaoQ on spatial asynchrony, respectively, while controlling the effects of all other variables included in the model (Table S3). Then we used multi-group SEM to test whether the pathways differ between IMP and QTP. Further, direct, indirect effects, and total effects of variables for the multi-group SEM were calculated by multiplying the standardized coefficients of the involved paths and summing all resulting products. To ensure that differences in sample size between the IMP and the QTP did not bias the results of our multi-group SEM, we conducted a sensitivity analysis by randomly selecting 999 QTP subsets matching the sample size of IMP and refitting the model for each subset. The original coefficients fall within the 95% confidence interval of the 999 resampled estimates, so we consider the results to be robust to differences in sample size (Figs. S12 and S13). The SEM was conducted using the psem function and multigroup function in the *piecewiseSEM* package[67]. The partial residuals from the SEM were estimated using the partialResid() function in the *piecewiseSEM* package[67]. All analyses were performed in R version 4.4.0[68]. To improve the normality, alpha stability, and spatial asynchrony were log-transformed.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

NDVI data were extracted from the NASA Landsat satellite mission via the GEE platform at earthengine.google.com. The temperature and precipitation data were extracted from NASA FLDAS[58] at disc.gsfc.nasa.gov/datasets/FLDAS_NOAH01_C_GL_M_001/summary?keywords=FLDAS. All aggregated data used in the analyses and visualization are available via Figshare, https://doi.org/10.6084/m9.figshare.27169641.

Code availability

The main R code used in this study is available at the figshare repository, https://doi.org/10.6084/m9.figshare.27169641.

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Author contributions

M.H., Y.H., and X.L. conceived and designed the study. X.L., Y.X., and M.H. collected the data. M.H. and Y.H. conducted statistical analyses with contributions from R.R.G. and D.M. All coauthors discussed the results and contributed to the final version. M.H. and Y.H. wrote the initial manuscript with inputs from all coauthors.

Competing interests

The authors declare no competing interests.

Additional information

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